700-km intercity Yb/Sr optical clock comparison at 2.3×10^{-18}

xxxxx

Abstract

Redefining the International System of Units (SI) second requires independent frequency-ratio measurements between clocks based on different atomic transitions in different laboratories, with an overall uncertainty below 5×10^{-18} and consistent results. However, the self-assessed uncertainties of these clocks have not yet been validated by such comparisons. We compared a ytterbium optical lattice clock in Wuhan with a strontium optical lattice clock in Shanghai—about 700 km apart—over a phase-stabilized fiber link spanning 1 350 km, and determine the Yb/Sr frequency ratio to be 1.207 507 039 343 337 720 1(28), with a Bayesian statistical uncertainty of 1.14×10^{-18} and a total uncertainty of 2.34×10^{-18}. The value differs from the NIST–JILA result 1.207 507 039 343 337 723 0(37) by -2.87×10^{-18}, consistent within the combined standard uncertainty. To our knowledge, this is the first intercity frequency-ratio measurement to meet the uncertainty requirement of the roadmap for the redefinition of the SI second, and together with the NIST–JILA result it provides the second, independent determination of the Yb/Sr ratio by a different institute, as required by criterion I.2.b for this pair of transitions. The measurement also demonstrates, at intercity scale, the capability that a wide-area optical-clock network requires.

Introduction

Optical lattice and single-ion clocks now reach fractional systematic uncertainties at or below 2×10^{-18} [1–8], two orders of magnitude better than current primary frequency standards[9;10]. Comparing independent clocks in different regions and laboratories verifies these self-assessed precision budgets more objectively: the environmental conditions and geodetic assumptions entering a self-evaluation are typically tied to a single location, and identical methods within one laboratory may share common systematic biases. Linking clocks of different types and locations into a collaborative network is therefore a key pathway toward establishing optical clocks as a verifiable metrological infrastructure[11;12]. The idea has moved from proposal to practice: fiber links with noise levels below those of the clocks themselves now span thousands of kilometers[13;14], and multi-clock campaigns have measured tens of ratios among clocks in six countries[15] and across a continental network[16]. Satellite-based advanced links with comparable performance[17–20] are also being developed and will interconnect with fiber networks to form a future global optical clock network.

The most immediate purpose of such a network is the redefinition of the SI second. One of the mandatory requirements (I.2.b) in the roadmap for this redefinition[21] is that at least five non-unit frequency ratios between standards of different clock transitions be measured, each ratio at least twice by different institutes, in agreement with an overall uncertainty of the comparison $\Delta\nu/\nu < 5 \times 10^{-18}$—a requirement reiterated in the 2025 CCTF progress report[22;23]. However, comparisons across independent institutions and different locations have reached at best a precision of 7.7×10^{-18} [16]. So far, no frequency-ratio measurements between clocks in different cities have satisfied this uncertainty requirement[24].

Closing that gap matters beyond the definition itself. Chronometrically leveled clocks map the gravitational potential of the Earth at centimeter resolution[1;25;26], and clock networks have been proposed and used as sensors for dark matter[27–29]. All of these applications assume that remote clocks agree at a level that, until now, has been demonstrated only locally, in three



Figure 1: The two-clock network. An IAPMST ytterbium clock in Wuhan and a USTC strontium clock in Shanghai are connected by a phase-stabilized 1 550 nm fiber link. Relay stations along the route carry erbium-doped fiber amplifiers and fiber-noise-cancellation loops.

experiments: a Sr/Yb/Al^+ triple-clock ratio measurement with an uncertainty of 3.2×10^{-18} by NIST–JILA in Boulder (US)[30], and two transportable-clock comparisons at the 10^{-18} level[26;31].

Here we report an intercity frequency-ratio measurement between an IAPMST ytterbium lattice clock in Wuhan and a USTC strontium lattice clock in Shanghai, about 700 km apart, connected by a 1 350 km fiber link. We determine the Yb/Sr ratio with a total uncertainty of 2.34×10^{-18}; gravitational redshift effects have been corrected, whereas measurement error and the time-varying gravitational redshift term associated with gravitational tides have been retained as uncertainty components. The result establishes the two sites as a prototype node pair for a distributed optical-clock network, and demonstrates, at intercity scale, the class of ratio measurement that the redefinition process depends on.

Results

0.1 Network and observation campaign

Figure 1 shows the two-clock network. A ytterbium (Yb) lattice clock at IAPMST[8] in Wuhan and a strontium (Sr) lattice clock at USTC[5] in Shanghai are separated by 700 km and compared through a 1 350 km phase-stabilized fiber link on a commercial telecom network used non-exclusively, with the carrier transmitted from Shanghai to Wuhan (Methods). Both combs are self-referenced and phase-locked to their local clock, so the two clocks are the system's only independent frequency sources; the recorded quantity is the out-of-loop beat between the received carrier and the local Yb-referenced comb, sampled every second and normalized to the carrier frequency f_{1550}, with the fiber loop-back monitored out of loop throughout. The geopotential difference between the two clocks was measured by spirit leveling to be $\Delta W = 280.042 \pm 0.085\ \mathrm{m^2\,s^{-2}}$ (Methods). Each observation window (segment) yields a frequency-ratio estimate through the fixed frequency-chain relations (Methods).

0.2 Clock-ratio determination

Over two months (58 days) the two clocks and their transfer chains ran synchronously for a total of 274 effective hours, recorded as 16 jump-free segments in three measurement stages defined

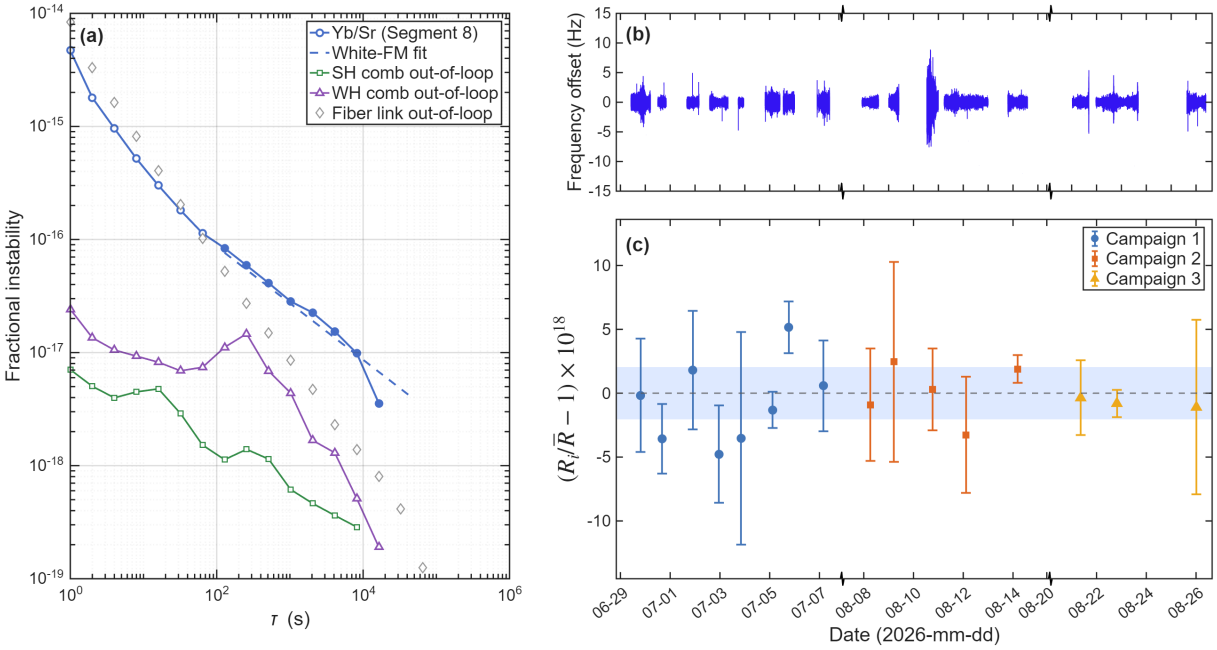


Figure 2: Frequency-ratio measurement results. (a) Fractional instability of a representative segment (Yb/Sr, segment 8); the gray markers are the out-of-loop instabilities of the two frequency combs (SH, WH) and of the fiber link. (b) Per-segment ratio deviations from the final reported value. (c) The same deviations by campaign date.

Table 1: Statistical combination of the 16 segment ratios. WLS denotes weighted least squares with inverse-variance weights $w_i = 1/u_i^2$; ξ is the inferred common excess scatter. χ^2_{red} is the reduced chi-square of the WLS combination (15 degrees of freedom, $p = 9.8 \times 10^{-1}$, Birge ratio 0.642).

Estimator	Yb/Sr	u (10^{-18})	ξ (10^{-18})	χ^2_{red}
WLS (inverse variance)	1.207 507 039 343 337 720 130	1.128	—	0.412
Mandel–Paule	1.207 507 039 343 337 720 130	1.128	0.00	1
Bayesian random effect	1.207 507 039 343 337 720 134	1.139	0.14	—

by the experiment team (late June to early July, early August, and late August); several of the segment boundaries correspond to deliberate clock shutdowns and restarts, so the reproducibility of the ratio rests on tested restart behavior rather than on continuous operation. The segment estimates show no systematic drift over the campaign, and the network tolerates interruption, provided each retained window is self-consistently characterized. The results are presented in Figure 2.

We determine the Yb/Sr frequency ratio from the 16 segments. Each segment yields a ratio estimate with its own statistical uncertainty (Methods). Following recent ratio comparisons[16;30], we combine the segments with a random-effect model that admits a common excess scatter[32;33], and evaluate three estimators (Table 1). The three agree at the 10^{-18} level; their center values spread by 3.7×10^{-21} in fractional units. We adopt the Bayesian random-effect estimate, which carries the largest standard uncertainty of the three, as the primary value,

$$\mathrm{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,1, \tag{1}$$

with statistical uncertainty 1.139×10^{-18}.

Table 2: Uncertainty budget of the Yb/Sr frequency-ratio measurement. Components are treated as independent; the fiber-link and frequency-comb entries are upper bounds (each below 1×10^{-19}, from the digitization error of the direct-digital-synthesis frequency-reference settings), carried at 0.1×10^{-18} in the quadrature.

Component	Contribution (10^{-18})
Statistical (Bayesian random effect)	1.139
Sr systematics	0.92
Yb systematics	1.3
Static gravitational potential	0.946
Fiber link	<0.1
Frequency comb	<0.1
Time-varying redshift	0.85
Total (quadrature)	2.34

0.3 Uncertainty budget

Beyond the statistical uncertainty of the ratio combination, the comparison carries several systematic contributions: the published evaluations of the two clocks, the geodetic determination of the gravitational-potential difference between the sites, and the residuals of the transfer subsystems. We discuss each in turn and combine them into the total standard uncertainty (Table 2).

The Sr clock contributes 0.92×10^{-18} from its published systematic evaluation[5], and the Yb clock 1.3×10^{-18}[8]. The static gravitational-potential term is the leveling uncertainty $0.085\ \mathrm{m^2\,s^{-2}}$ converted to fractional frequency as $u(\Delta W)/c^2$, giving 0.946×10^{-18}; this is the same leveled geopotential difference already applied as the static correction (Section 0.1). The frequency conversion stages each contribute below 1×10^{-19} from the digitization error of the direct-digital-synthesis frequency-reference settings, carried as 0.1×10^{-18} and 0.1×10^{-18} upper bounds in the budget.

The time-varying gravitational redshift is more involved. Solid Earth and ocean tides, together with atmospheric, hydrological and non-tidal ocean loading, move the local potential by amounts reaching the 10^{-18} level over the campaign[34–36]. Correcting the beat record with a tide template from the public software ETideLoad4.5[37], which implements the IERS Conventions (2010)[38], degrades rather than improves the fit, raising the reduced chi-square of the combination from 0.41 to 0.54, and a second, independent template built from the DE421 ephemerides[39] and the FES2022 ocean-tide model[40] does the same ($\chi^2_{\mathrm{red}} = 0.50$) while moving the center in the opposite direction, by -0.85×10^{-18} versus $+0.21 \times 10^{-18}$ (Methods). The sign disagreement between the two models confirms that the templates are incomplete at this level, so the time-varying term is carried as an uncertainty rather than a correction, at the larger of the two model shifts, 0.85×10^{-18} (Methods).

Combining the statistical and systematic components in quadrature gives the total standard uncertainty 2.34×10^{-18}; the reported ratio is therefore

$$\mathrm{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,1(28), \tag{2}$$

where the parenthesized digits are the total standard uncertainty in the last two decimal places of the value.

Discussion and outlook

The value reported here agrees with the NIST–JILA measurement at the level set by the redefinition roadmap (Figure 3): relative to the NIST–JILA value $1.207\,507\,039\,343\,337\,723\,0(37)$[30]

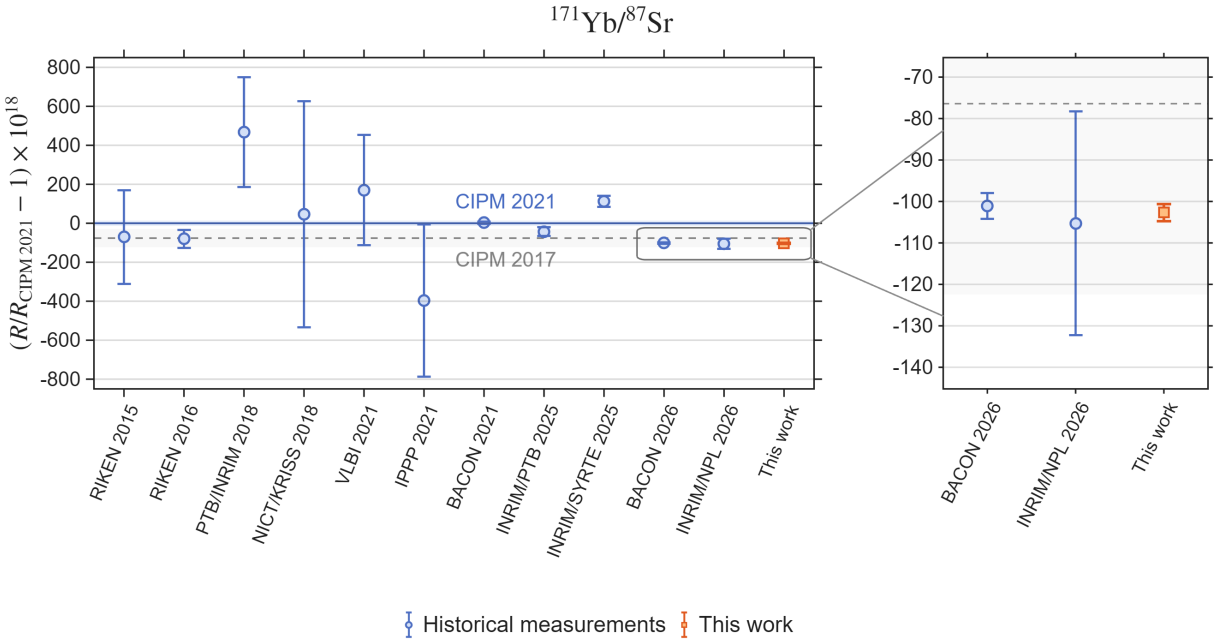


Figure 3: Evolution of the Yb/Sr frequency-ratio measurements. Deviations from the final value are shown against the measurement year for the historical determinations and for this work.

our central value differs by -2.87×10^{-18}, well within the 5×10^{-18} agreement threshold of the roadmap's frequency-ratio criterion[21;23], and within the combined standard uncertainty of the two measurements. The European-network value 1.207 507 039 343 337 718(32)[16], which differs from ours by $+2.13 \times 10^{-18}$, provides a supporting cross-check at a larger uncertainty of 3.2×10^{-17}.

The evolution of the Yb/Sr ratio also shows how comparison measurements drive the field forward. Since 2015[41], independent ratios were measured with increasing precision, yet the determinations disagreed well beyond their quoted uncertainties—an early sign of common-mode errors in the absolute frequency chain on which the earlier values relied[29]. The successive comparisons exposed those errors and guided their elimination (Figure 3).

Changes in the Earth's gravitational potential enter the comparison directly. The static difference between the two clock transitions is fixed by first-order spirit leveling and applied as a correction, although it rests on a single leveling campaign whose available GNSS-plus-geoid-model calibration is too imprecise (Methods) to act as an effective cross-check; closing this gap will require transportable optical clocks with higher frequency reproducibility, rather than geodetic techniques alone.

The time-varying part is observed directly. In any single segment the 1200-s demeaned beat waveform and the tidal template are too noisy to resolve, but across the 16 segments entering the ratio combination the combined evidence is decisive: the sign of the within-segment correlation is positive in 12 out of 16 segments for both tide models, and the precision-weighted response amplitude differs from zero at 5.0σ for the ETideLoad4.5 template (+0.58) and at 5.7σ for the DE421+FES2022 template (+0.87) (Methods). A signal at the tidal period is therefore present in the record at the 10^{-18} level, yet no template captures it: the two amplitudes differ from each other and both fall short of the full expected response of +1, and the two models disagree in the sign of the center shift they produce. The observed coupling is smaller than the templates predict, so carrying the larger model shift as an uncertainty item (Section 0.3) is conservative; a tide model that correctly describes the data would reduce this term, but constructing one requires more comparison data and dedicated geodetic inputs, which is beyond the scope of this paper.

The effect grows with the scale of the comparison: on longer, continental- and intercontinental-scale baselines the time-varying redshift becomes one of the dominant systematic terms[36]. That same sensitivity turns the network into an instrument for the Earth: clock comparisons over large baselines directly sense the time-variable gravitational potential, providing an independent means of testing and calibrating tide and loading models at the 10^{-18} level—the model disagreement seen here can be refined, or removed, by more comparison data from an expanding optical-clock network. Conversely, tidal knowledge gained in this way feeds back into the comparisons themselves: with data-consistent models, the time-varying term can be corrected rather than carried, extending the accuracy reach of large-scale clock comparisons.

Looking ahead, the result is a milestone on the necessary path toward a wide-area optical-clock network: the building blocks are now in place—terrestrial fiber links of the kind used here[13;14] carry comparisons over thousands of kilometers within continents, while free-space and satellite-based links are closing the remaining gaps[17–19;42]. The natural next step is a hybrid architecture in which optical-clock nodes on continental fiber backbones are interconnected across oceans by satellite or free-space optical links, so that any pair of clocks on Earth can be compared at the level demonstrated here. Such a network and the geodetic infrastructure will develop together: precise geopotential referencing supports the comparisons, while the network's time-variable gravity observations in turn refine the geodetic models[1;25].

Conclusion

Together with the NIST–JILA measurement, this comparison provides, for the first time, two independent determinations of the Yb/Sr ratio in agreement at the precision required by the roadmap for the redefinition of the SI second, satisfying criterion I.2.b of the roadmap for this pair of transitions. The two sites now form a working node pair of a distributed optical-clock network: with more optical-clock nodes joined through terrestrial fiber backbones and interconnected across continents by satellite and free-space links, the comparison demonstrated here extends naturally into the global network on which the redefinition, chronometric leveling of the gravitational potential, and distributed searches for new physics will depend.

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Methods

Leveling determination of the geopotential difference

The static gravitational-redshift correction is fixed by the geopotential difference between the atomic-transition positions of the two clocks, determined by a first-order spirit leveling campaign carried out specifically for this comparison: a 1 113 km line between the Wuhan benchmark WS-03 and the Shanghai benchmark WS-01, observed in both directions with digital levels and barcode invar staffs, with gravity observed along the line. The geopotential difference is integrated section by section, $\Delta W_{AB} = -\sum_i g_i \, \Delta H_i$, and combined with the local benchmark-to-transition offsets, giving $\Delta W = 280.042 \pm 0.085\ \mathrm{m^2\,s^{-2}}$, equivalent to an optical fractional shift of $-3.116 \times$

10^{-15}. The per-kilometer accidental error $M_\Delta = 0.255\,\mathrm{mm\,km^{-1}}$ satisfies the most stringent first-order requirements of the Chinese, US, German and Czech national standards (Supplementary Section 3)[43–46]. An independent GNSS-plus-EGM2008 geoid-model calibration gives $280.60 \pm 2.558\ \mathrm{m^2\,s^{-2}}$, consistent with the leveled value but about 30 times less precise, so it serves as a consistency check rather than a calibration at the reported accuracy; a meaningful independent verification will require transportable optical clocks with higher frequency reproducibility.

Experimental setup and frequency transfer

The two sites are connected by a phase-stabilized fiber link of 1 350 km single-pass length (2 700 km loop-back), operated at 1 550 nm and carried on a commercial telecom network used non-exclusively. The route is divided into spans of tens of kilometers, each terminated by a relay station carrying a narrow-linewidth laser, an erbium-doped fiber amplifier and a fiber-noise-cancellation loop; the added instability of the link is kept below the clock noise for averaging times beyond 100 s.

Both ends use independent optical frequency combs, self-referenced and phase-locked to their local clock, to bridge the spectral gap between the clock transitions and the telecom carrier; all frequencies in the chain derive from the two clock transitions through fixed factor-of-two relations, $f^{1397}_{\mathrm{comb}} = \frac{1}{2}(1+a)\,f_{\mathrm{Sr}}$ and $f^{1156}_{\mathrm{comb}} = \frac{1}{2}(1+b)\,f_{\mathrm{Yb}}$, where a and b collect the Sr and Yb systematic corrections (density, lattice, second-order Zeeman and blackbody shifts). The observable is the out-of-loop beat between the received carrier and the local Yb-referenced comb, converted to fractional units with the transfer frequency $f_{1550} = 193.3992\,\mathrm{THz}$. The static gravitational-redshift correction enters the ratio inversion as a multiplicative factor, $(\mathrm{Yb/Sr})_{\mathrm{corr}} = (\mathrm{Yb/Sr})_{\mathrm{raw}} \times (1+\delta_g)$ with $\delta_g = -3.116 \times 10^{-15}$, derived from the leveled geopotential difference above.

Data processing and segment definition

The counters sample the beat every second in phase-averaging mode. Each accepted segment contributes its longest jump-free record. Supplementary Table 1 lists the per-segment retained counts, ratio deviations and statistical uncertainties; five segments whose stability slopes fall outside the white-frequency-noise band are flagged there as model-limited. These five lie at the steep end of a continuous slope distribution, just beyond the acceptance bound, and show no common campaign, weather or laboratory condition that distinguishes them from the remaining segments.

As an additional stability check, the 16 segments were spliced end-to-end on a continuous 1-s clock with the wall-clock gaps removed, giving a single 986 416 sample record whose overlapping Allan deviation $\sigma_y(\tau)$ can reach averaging times beyond a single segment (Figure 4). The spliced curve extends the per-segment stability points to τ up to about 4.6 days of pure data time, at which the raw record reaches $\sigma_y \approx 8.6 \times 10^{-19}$; applying the two theoretical tide corrections ($A = +1$) shifts the long-τ level by less than 10% in opposite senses, and the removed correction terms (h/f_{1550}) themselves lie below the corrected curves. Because the spliced axis removes the inter-segment gaps, points with τ exceeding one segment mix different campaigns; the pair count behind each point is reported with the data. This diagnostic is a data record, not a statistical uncertainty.

Clock-ratio determination, statistical model and uncertainty budget

Each retained segment yields a clock-ratio estimate R_i from the mean of its retained beat samples, inverted through the fixed frequency-chain relations and corrected for the static gravitational redshift $\delta_g = -3.116 \times 10^{-15}$ derived from the leveled geopotential difference (see above). The per-segment statistical uncertainty u_i is obtained from the overlapping Allan deviation of the beat fractional frequency, evaluated on a tau grid of powers of two plus 600, 1200, 3600 and

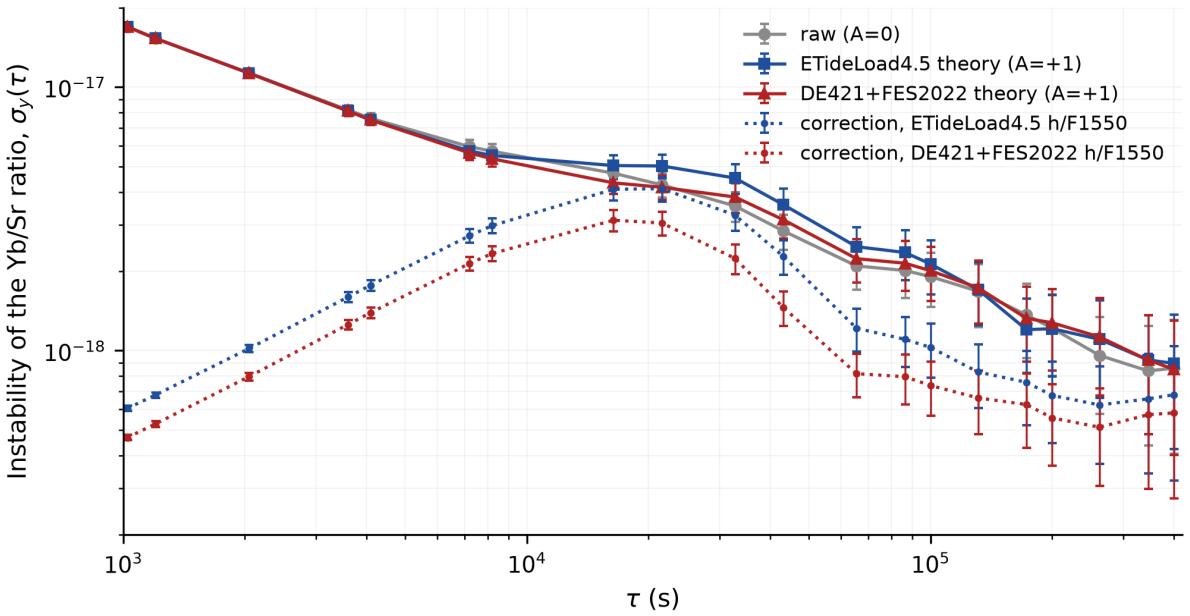


Figure 4: Spliced concatenated stability of the 16 segments. The segments are placed end-to-end on a continuous 1-s clock (gaps removed, 986 416 samples, 11.4 d of pure data time); one overlapping Allan deviation per series, from $\tau = 1\,000$ s to 4.6 d. Curves: the raw record ($A = 0$), the theoretical tide correction ($A = +1$) for the ETideLoad4.5 and DE421+FES2022 templates, and the removed correction terms h/f_{1550} themselves (dotted). Error bars are 1-σ equivalent degrees of freedom; the pair count at each τ is reported with the data. Points beyond one segment mix campaigns, so the curve is a data record rather than an uncertainty estimate.

7200 s, all limited to a quarter of the longest gap-free run T_i of the segment: u_i adopts the last computable point $u_i = R_i\,\sigma_y(\tau_{\max})$ without extrapolating the stability curve beyond the data. The log-log slope of σ_y against τ is computed as a diagnostic; segments whose slope leaves the white-frequency-noise band are flagged but retained.

The segment estimates are then combined under a random-effect model with a common excess scatter[32;33]: each segment is modeled as $R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2)$, with ξ added in quadrature to the per-segment uncertainties. The primary value is obtained from a Bayesian random-effect model, in which μ and ξ are jointly inferred with flat priors on μ and on $\log\xi$, and the joint posterior $p(\mu, \xi \mid \{R_i, u_i\})$ is marginalized numerically over a 40-decade $\log\xi$ grid, with the grid span expanded until the posterior mass at the edges falls below 1%; the posterior mean of μ and its posterior standard deviation give the reported central value and statistical uncertainty, with $\xi = 0.14 \times 10^{-18}$. As cross-checks, the same observation model is combined by weighted least squares (inverse-variance weights $w_i = 1/u_i^2$) and by the Mandel–Paule estimator[33]; the resulting χ^2_{red} and ξ values are given in Table 1.

The remaining terms of the uncertainty budget (Table 2) are systematic. The Sr and Yb terms are the published evaluations of the two clocks[5;8]; the value adopted for the Yb clock is the published evaluation of 1.3×10^{-18} [8], which is more conservative than an internal estimate, and the discrepancy does not affect the conclusion at the reported precision. The static-potential term is $u(\Delta W)/c^2$ from the leveling uncertainty 0.085 $\mathrm{m^2\,s^{-2}}$, giving 0.946×10^{-18}. The frequency-conversion stages contribute below 1×10^{-19} each, from the digitization error of the direct-digital-synthesis frequency-reference settings; the budget carries them as 0.1×10^{-18} upper bounds.

Tidal template and time-varying redshift

The time-varying gravitational-redshift term is quantified from a tidal template: the solid Earth tide and ocean tide loading displacement difference between the two sites, computed every 30 s with the public software ETideLoad4.5, converted to an equivalent geopotential difference and then to a beat-frequency template $h = f_{1550}\,\Delta W/c^2$ with the same f_{1550} normalization as the beat record. A second, independent template is built from the DE421 ephemerides and the FES2022 ocean-tide model on a 90 s grid.

To test how well the templates fit the data, we repeated the ratio combination with the tide series applied before the inversion, at two fixed response amplitudes: the theoretical amplitude $A = +1$ and the historical amplitude $A = +0.54$ from the demeaned beat waveform of our earlier analysis of the same campaign. Both templates degrade rather than improve the fit: the reduced chi-square of the combination rises from 0.41 to 0.57 and 0.54 for the ETideLoad4.5 template (Bayesian center shift -0.85×10^{-18}), and to 0.50 for the DE421+FES2022 template, which moves the center in the opposite direction, by $+0.21 \times 10^{-18}$. Repeating the segment-mean correlation with the solid tide alone ($r = +0.36$ instead of $+0.43$) locates the dominant coupling in the ocean-tide loading term.

Templates that degrade the segment agreement, and whose two independent realizations disagree in sign, are treated as inconsistent with the data: no tidal correction is applied to the reported central value, and the larger of the two center shifts is carried as the time-varying-redshift uncertainty item 0.85×10^{-18}.

Within-segment tidal correlation analysis

The within-segment correlation analysis uses the same 1200-s triangular window as the ratio processing: the retained beat record of each segment and the 1-s tidal template are both window-averaged with a 1200-s triangular window at 600-s stride, demeaned, and fitted through the origin as beat $= A \cdot$ tide, with the template sign corrected for the beat polarity so that a positive amplitude corresponds to the expected response $A = +1$; the amplitude, the Pearson correlation r and its p value are inferred from the non-overlapping 1200-s windows only, with the per-segment amplitude uncertainty obtained from the scatter of the non-overlapping residuals. No single segment reaches significance on its own; across the 16 segments entering the ratio combination, the sign of the correlation is positive in 12 of the 16 segments for both tide models, and the precision-weighted amplitudes are +0.58 (ETideLoad4.5) and +0.87 (DE421+FES2022), both more than 5σ from zero yet short of the expected +1 and differing from each other—the same conclusion as the spliced stability record above. Because the two models are correlated realizations that share the same beat record, their two significance values are not independent replications of the measurement. Surface air temperature records at the sites show a clear diurnal cycle (peak-to-peak ∼4.6–5.6 °C), but the diurnal components of the out-of-loop reference, the out-of-loop signal and the fiber-link beat are each far below their respective measurement noise, and the beat reflects the frequency difference between the two clocks, so the temperature cycle is not detectable in the record and does not drive the tidal-period correlation. The observed coupling is smaller than the templates predict, and the tidal term is kept as an uncertainty rather than a correction, as in the ratio combination above.

The components are treated as independent and combined in quadrature, giving the total standard uncertainty 2.34×10^{-18}.

Data availability

The frozen, machine-readable result layer underlying this manuscript is archived in the repository (`https://github.com/hjiang555-a11y/gravitation-whsh`) under `results/audit-v1/verified/`: the manifest with SHA-256 `ac725aaba8ec744bf4e7a17cf37cee41af57461f0d3684584711b24a02b0962e`,

the derived ledgers and statistics, and the generated audit report. The raw beat files and the leveled geopotential inputs are available from the corresponding author upon reasonable request.

Code availability

The analysis and manuscript pipelines are in the same repository under the Apache License 2.0. The manuscript numbers are generated by `paper/rev2/tools/build_numbers.py`, which reads only the frozen manifest and the curated constants listed in `generated/numbers_manifest.json`.